

Surfaces. Spherical and cylindrical coordinates.

[SUBMIT ANSWERS TO ALL QUESTIONS]

A surface can be defined implicitly by an equation of the form $f(x, y, z) = C$, or given in parametric form: $x = x(u, v)$, $y = y(u, v)$, $z = z(u, v)$ [i.e., $\mathbf{r} = \mathbf{r}(u, v)$].

A vector $\mathbf{n}$ normal to the surface can be found as ∇f or $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$.

The equation of the *tangent plane* to a surface at point $\mathbf{r}_0$ is $(\mathbf{r} - \mathbf{r}_0) \cdot \mathbf{n} = 0$.

Spherical coordinates: radius r , polar angle θ , azimuth ϕ ($0 \leq r < \infty$, $0 \leq \theta \leq \pi$, $0 \leq \phi \leq 2\pi$),

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta,$$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \cos \theta = z / \sqrt{x^2 + y^2 + z^2}, \quad \tan \phi = y/x, \quad .$$

Length element (squared), spherical surface element and volume element:

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2, \quad dS = r^2 \sin \theta d\theta d\phi, \quad dV = r^2 \sin \theta dr d\theta d\phi.$$

Cylindrical coordinates: radius ρ , angle ϕ , z ($0 \leq \rho < \infty$, $0 \leq \phi \leq 2\pi$, $-\infty < z < \infty$),

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z,$$

$$\rho = \sqrt{x^2 + y^2}, \quad \tan \phi = y/x, \quad z = z.$$

Length element (squared), cylindrical surface element and volume element:

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2, \quad dS = \rho d\phi dz, \quad dV = \rho d\rho d\phi dz.$$

1. A parabolic surface (paraboloid of revolution) is described by the equation $z = 1 - x^2 - y^2$.

(a) Verify that point $\mathbf{r}_0 = (1, 1, -1)$ lies on this surface. Write the equation of the surface in the form $f(x, y, z) = 1$ and find the normal $\mathbf{n} = \nabla f$ to the surface at point $\mathbf{r}_0$. Hence, find the equation of the tangent plane at this point. [15]

(b) Show that this surface can be described in parametric form by

$$\mathbf{r}(u, v) = u\mathbf{i} + v\mathbf{j} + (1 - u^2 - v^2)\mathbf{k}.$$

Verify that $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$ is the same vector as $\mathbf{n}$ in part (a) at any point on the surface. [15]

2. (a) Show that the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, can be parameterised as [10]

$$\mathbf{r} = a \sin u \cos v \mathbf{i} + b \sin u \sin v \mathbf{j} + c \cos u \mathbf{k} \quad (0 \leq u \leq \pi, \quad 0 \leq v \leq 2\pi).$$

(b) Show that $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}$ is collinear to $\mathbf{n} = \nabla f$, where $f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}$. [10]

3. For two points in Cartesian coordinates, $A(1, 2, 2)$ and $B(-1, -1, -2)$, find their cylindrical coordinates (ρ, ϕ, z) and spherical coordinates (r, θ, ϕ) . [20]

4. Using cylindrical coordinates, the position vector can be written as

$$\mathbf{r} = \rho \cos \phi \mathbf{i} + \rho \sin \phi \mathbf{j} + z \mathbf{k}.$$

Find vectors $(d\mathbf{r})_\rho = \frac{\partial \mathbf{r}}{\partial \rho} d\rho$, $(d\mathbf{r})_\phi = \frac{\partial \mathbf{r}}{\partial \phi} d\phi$ and $(d\mathbf{r})_z = \frac{\partial \mathbf{r}}{\partial z} dz$, and show that the volume of the parallelepiped formed by vectors $(d\mathbf{r})_\rho$, $(d\mathbf{r})_\phi$ and $(d\mathbf{r})_z$ is $dV = \rho d\rho d\phi dz$. [30]

[Hint: this volume can be found as $(d\mathbf{r})_\rho \cdot [(d\mathbf{r})_\phi \times (d\mathbf{r})_z]$.]